

Dynamics 365 Finance & Operations (D365 FO)

Master Planning and Action Messages – Detailed Functional Overview

Introduction

Master Planning in Dynamics 365 Finance and Operations is a core Supply Chain Management capability that helps organizations plan procurement, production, and inventory based on actual demand, forecasts, and supply parameters. It ensures material availability at the right time while optimizing inventory levels and operational costs.

This document provides a professional and structured explanation of **Coverage Groups**, **Master Plans**, **Net Requirements**, and **Action Messages**, focusing on how these components work together to generate and manage planned orders.

Coverage Groups

Coverage Groups are one of the most critical configurations in Master Planning. They define *how* and *when* demand is covered for items by controlling time fences, planning logic, and replenishment strategies. Once defined, coverage groups are assigned to items and directly influence planned order generation.

Standard view ▾

Coverage groups

Coverage group	Name	Calendar
20Days	20 Days	24h

General

COVERAGE

Coverage code

Period

Coverage period

Coverage time fence (days)

365

Negative days

2

Positive days

365

Planning priority model

PEGGING SEQUENCE

Consume on-hand inventory

Before all other supply

Use latest possible supply

OTHER

Use the specified BOM or formul...

Prioritize existing supply over re...

Strict safety stock pegging

Other

PLANNED ORDER

Requested production status

Scheduled

Automatic firming time fence (da...

Freeze time fence (days)

BOM explosion time fence (days)

200

Capacity scheduling time fence (...)

200

REQUISITION

Approved requisitions time fence...

0

FORECAST PLAN

Forecast plan time fence

100

Reduction key

Reduce forecast by

Orders

Include intercompany orders

Include customer forecast in the ...

SAFETY MARGINS IN DAYS

Receipt margin added to require...

0

Issue margin deducted from req...

0

Reorder margin added to item le...

0

DDMRP PARAMETERS

Lead time factor

1.00

Variability factor

1.00

Min, max, and reorder-point per...

Daily

AVERAGE DAILY USAGE

Average daily usage based on

Past

Past period (days)

10

Forward period (days)

10

Relative weight of past period fo...

50.00

Relative weight of forward perio...

50.00

Forecast model

Action

ACTION MESSAGE

Action message

No

Action time fence

100

Postpone margin

0

Advance margin

0

Round down advance action days

No

Basis date

Requirement date

Advance

Yes

Postpone

Yes

Decrease

Yes

Increase

Yes

Derived actions

Yes

Coverage Code Options

1. Period

The *Period* coverage code is used when demand is planned within defined time intervals rather than individually per requirement.

Key characteristics:

- Demand within the specified period is grouped together.
- Planned orders are generated based on the defined coverage period.
- Suitable for items with stable or predictable demand patterns.

Important parameters:

- Coverage period (days, weeks, or months)
- Safety margin consideration
- Forecast aggregation within the period

This method reduces the number of planned orders and supports smoother procurement or production cycles.

2. Requirement

The *Requirement* coverage code creates a planned order for each individual demand.

Key characteristics:

- One-to-one relationship between demand and supply.
- Each sales order, forecast line, or requirement generates its own planned order.
- High accuracy but may result in a larger number of planned orders.

This option is best suited for:

- High-value items
 - Make-to-order scenarios
 - Items with irregular or customer-specific demand
-

3. Min / Max

The *Min/Max* coverage code is inventory-driven rather than demand-driven.

Key characteristics:

- Inventory levels are maintained between minimum and maximum quantities.
- When inventory falls below the minimum, master planning generates a replenishment order up to the maximum level.
- Demand dates are less critical than stock levels.

This option is commonly used for:

- Consumables
 - Fast-moving spare parts
 - Items requiring continuous availability
-

4. Manual

The *Manual* coverage code excludes the item from automatic master planning.

Key characteristics:

- No planned orders are automatically created.
- Planner manages replenishment manually.
- Useful for exceptional or non-standard items.

This option is often used when planning decisions require direct human control.

5. Priority

The *Priority* coverage code allows multiple coverage groups to be evaluated in a priority sequence.

Key characteristics:

- Coverage logic is applied based on priority order.
- Enables advanced planning strategies.
- Supports complex business scenarios where different planning rules apply under different conditions.

Master Plans

After defining coverage groups, the next step is to configure **Master Plans**. Master Plans control how master planning is executed and which parameters are applied during calculation.

Each master plan can be designed for a specific purpose, such as static planning, dynamic planning, or simulation.

Standard view ▾

Master plans

Plan	Name
App-Plan	Apparel Master Plan

General

SETUP

Reserved to ComptE service
☒ No

Exclude stopped orders
☒ No

Include on-hand inventory
☒ No

Override pegging sequence
☒ No

Consume on-hand inventory
Before all other supply

Use latest possible supply
☒ No

Include inventory transactions
☒ Yes

Include sales quotations
☒ No

Probability %
0

Include requests for quotations
☒ No

Include requisitions
☒ No

Use shelf life dates
☒ No

Include sales orders
☒ Yes

Include transfer orders
☒ No

Include purchase returns
☒ Yes

Include retail sales orders
☒ Yes

Include continuity plan
☒ No

Exclude specific transactions
☒ No

Exclude items without on-order ...
☒ No

Autofirm orders even with errors
☒ No

Calculate Batch CTP even with er...
☒ No

PLANNED PRODUCTION ORDERS

Scheduling method
Operations scheduling ▾

Finite property
☒ No

Backward scheduling capacity ti...
180

Finite capacity
☒ No

Finite capacity time fence
365

Finite capacity for bottleneck res...
☒ No

Capacity time fence for bottlenec...
0

FORECAST

Forecast model
▾

Include supply forecast
☒ No

Include demand forecast
☒ No

Method used to reduce forecast ...
None ▾

NUMBER SEQUENCES

Planned orders
Mast_1 ▾

Session
Mast_3 ▾

CONTINUITY

Continuity plan
▾

Time fences in days

Coverage ☒ No 100	Firming ☒ No 0	Forecast plan ☒ No 100	Capacity ☒ Yes 365	Action message ☒ No 80	Approved requisitions time fence... ☒ No 0	Negative days ☒ No 0
Freeze ☒ No 0	Explosion ☒ Yes 365	Exclude demand forecast for nex... ☒ No 0	Continuity plan ☒ No 0	Calculated delays ☒ No 100	Sequencing ☒ No 0	Positive days ☒ No 0

Calculated delays

Ensure that the planned orders a... ☒ Yes	PLANNED PURCHASE ORDERS Add the calculated delay to the r... ☒ No	PLANNED PRODUCTION ORDERS Add the calculated delay to the r... ☒ No	PLANNED TRANSFER Add the calculated delay to the r... ☒ No	PLANNED KANBAN Add the calculated delay to the r... ☒ No	JUST IN TIME Just in time plan for the compon... ☒ No Just in time minimum days 1
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General Tab Parameters

The General tab defines the overall behavior of the master plan.

Key parameters include:

- Plan type (Static or Dynamic)
- Coverage group selection
- Forecast plan association
- Inclusion of on-hand inventory, purchase orders, and production orders

These settings determine the data scope and logic used during master planning runs.

Time Fence Parameters

Time fences define how master planning treats demand and supply within specific time horizons.

Key time fences include:

- Freeze time fence: Prevents changes to planned orders within a fixed period
- Firm time fence: Planned orders inside this fence are automatically firmed
- Forecast time fence: Determines how forecast consumption behaves

Time fences provide stability and control, ensuring that short-term plans are not constantly changing.

Calculated Parameters

The Calculated tab controls performance and calculation behavior.

Key parameters include:

- Explosion level
- Regenerative or net change planning
- Caching and update settings

These parameters influence calculation speed, system performance, and planning accuracy.

Assigning Coverage Groups to Items

0.00 Serial number group es Record the attribute value Receipt entry Latch weight item No Automatic GTIN No

Engineer es BOM

BOM unit es	Costing level 0	Phantom No	Width	CALCULATION Calculation group STD	Production group	Flushing principle Start	FORMULA PLANNING Production type BOM
Constant scrap	Planning level 0	Auto-report as finished No	Depth	PRODUCTION Production pool LACL	Property	Release to warehouse Allow partial reservation	Planning formula
Variable scrap	Cost calculation level 0	MEASUREMENT Height	Density		Arrival No		

Plan REQ

Coverage group
REQ

Manage costs FN 0.00

POSTING Item group FN	COSTING Cost group FN Use cost price by variant Yes	PRICE UPDATE Latest cost price Yes Date of price	PRICES Unit es	Price 0.00 Price quantity 1.00	CHARGES Price charges 0.00	Charges quantity 0.00 Ind. in unit price No	ABC CLASSIFICATION Value None Margin None Revenue None Carrying cost None
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Once coverage groups and master plans are configured, coverage groups are assigned to items.

Steps:

- Open the released product
- Navigate to the Plan tab
- Assign the appropriate coverage group

This assignment ensures that master planning applies the correct planning logic for each item.

Net Requirements and Planned Orders

Product information management > Products > Released products

← Edit + New Delete Product Purchase Sell Manage inventory Engineer **Plan** Manage projects Manage costs Commerce Gener

View Supply schedule	Forecast Demand forecast Supply forecast Inventory forecast	Coverage Item coverage	Order settings Default order settings	Requirement Net requirements Gross requirement	Sequencing Sequence value
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Released product details | Standard view

After configuration, planners calculate demand and supply using **Net Requirements**.

Process:

- Open the released product
- Go to the Plan tab
- Select Net requirements
- Run the Update function

The system calculates:

- Planned purchase orders
- Planned production orders
- Planned transfer orders

These planned orders are generated based on demand, coverage settings, and master plan parameters.

Firming Planned Orders

Once planned orders are reviewed, they are firming to convert them into actual execution documents.

Examples:

- Planned purchase orders are firming into purchase orders
- Planned production orders are firming into production orders

Firming marks the transition from planning to execution and allows operational teams to proceed with procurement or production.

Action Messages

Action Messages are system-generated recommendations that guide planners in managing changes and exceptions.

Common action messages include:

- Expedite: Suggests moving an order earlier
- Postpone: Suggests delaying an order
- Cancel: Indicates unnecessary orders
- Increase or Decrease quantity

Action messages help planners react proactively to demand changes, supply delays, and planning conflicts while maintaining optimal inventory levels.

Conclusion

Master Planning in Dynamics 365 Finance and Operations is a powerful engine that integrates demand, supply, and inventory planning. By correctly configuring coverage groups, master

plans, and time fences, organizations can achieve accurate planning, reduce costs, and improve service levels.

Action Messages further enhance decision-making by providing clear and actionable insights, enabling planners to respond effectively to changing business needs.

This structured approach ensures a smooth transition from demand planning to execution while maintaining operational control and planning stability.